DIRECTED ENERGY WEAPONS: Defining Future Warfare

What if a weapon did not need ammunition? Directed energy systems are bringing that question closer to reality while creating an entirely new set of engineering challenges.



DR DEEPAK HALAN is Professor with Jaipuria Institute of Management, NOIDA.

"Directed energy weapons (DEWs) technologies are essential since they provide an effective mechanism for contactless, non-kinetic means to achieve superiority in space. DEWs are fast emerging as an alternative to direct ascent ASAT missiles and are difficult to attribute to a source."

—Manohar Parrikar Institute for Defence Studies and Analyses, India

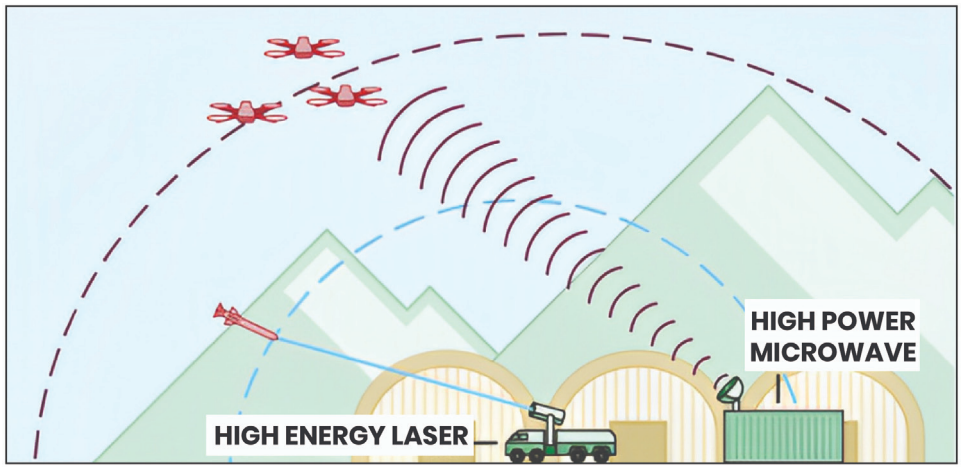
Directed energy weapons (DEWs) are electromagnetic systems capable of converting chemical or electrical energy into radiated energy, such as lasers, microwaves, particle beams, or even acoustic beams, and focusing them on a target to cause serious damage. DEWs are a revolutionary advancement in military technology, capable of degrading, incapacitating, or destroying enemy assets without relying on conventional ammunition. They deliver at the speed of light, offer precision, scalability, and reduced logistical burden.

Although research has been underway for decades, the progress has been slow due to technical hurdles. Today, DEWs are viewed as futuristic systems suiting the regional security environments. Across the world, strategists are looking at these weapons as next-generation systems for air and missile defence, counter-drone operations, and protection against asymmetric threats. Beyond the battlefield, directed energy technologies hold promise in space security and industrial use. High-energy lasers find use in precision cutting, welding, and surface treatment. Particle beams are used for targeted medical treatments, sterilising food and water, and for long-distance communication in space. India needs to reduce dependence on other countries and develop an indigenous ecosystem for the next-generation DEWs. Various stakeholders such as DRDO labs, electronic component and defence equipment manufacturers, and startups and academia will need to work closely together.

DEWs: Types

DEWs use the principle of electromagnetics (see Box 1) and are categorised according to their energy source. High-energy lasers, high-power microwaves, particle beams, and acoustic/hybrid systems each represent a distinct technological approach. Their operational advantages (Table 1) and technical characteristics (Table 2) vary significantly.

High-energy lasers (HELs). HELs use concentrated beams of light to damage targets using intense heat. Unlike conventional bullets, laser beams travelling at the speed of light hit fast-moving threats like missiles or drones almost instantly. The beam stays focused for a few seconds to burn through the target's outer shell or to disable sensors.



DEWs are categorised as high-energy lasers or high-power microwaves (Source: https://visionias.in)